

TEAM CHANDRA

Urban Heat Island Mitigation Simulation

UHIMS

AI-Driven Urban Heat Island Mitigation Simulator

Presented By

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THE PROBLEM: URBAN HEAT ISLANDS



The Core Issue

Dense urban areas trap heat due to concrete surface, roads, and a lack of green cover, creating dangerous temperature spikes.



The Planning Gap

Urban planners lack localized, data-driven tools to test how much a new park, cool roof, or water body will lower the temperature before physical construction.



THE IDEA

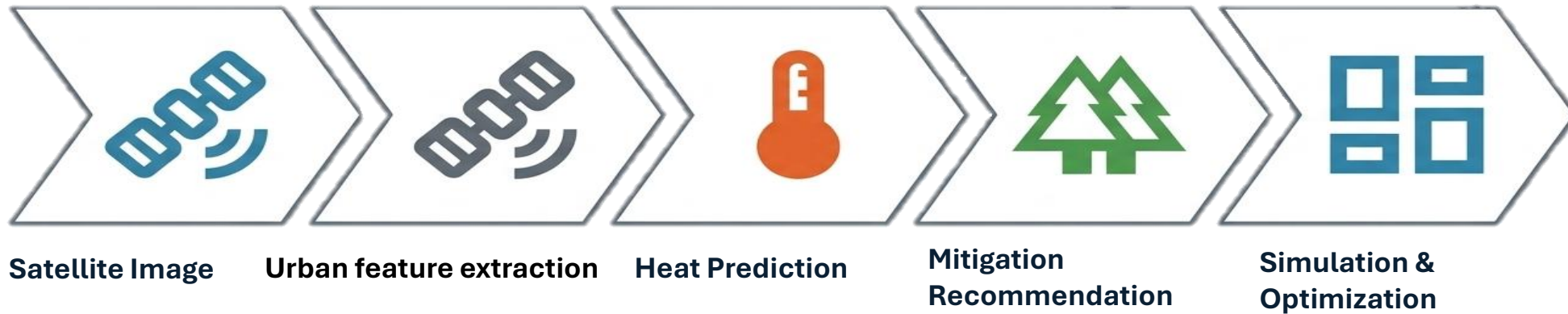
THE PROBLEM

- High temperatures from buildings, roads & low vegetation.
- Dense urban areas trap more heat.
- Need localized, data-driven heat analysis.
- Need to test mitigation strategies before implementation.

OUR PROPOSED SOLUTION

- UHIMS
 - ❖ AI-Driven Urban Heat Island Mitigation Simulator
 - ❖ AI + Computer Vision + Geospatial Analysis + Simulation

CORE PIPELINE



THE INNOVATION

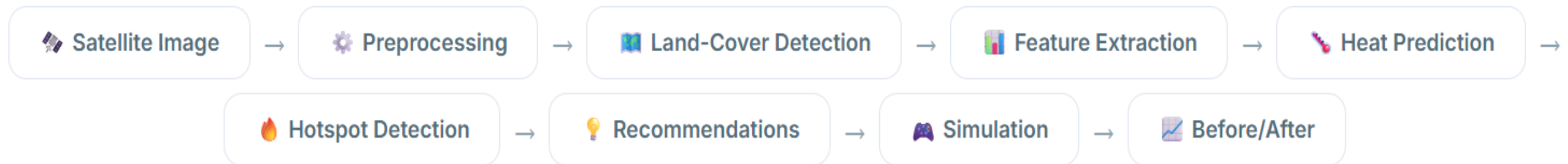
- Detects heat sources and urban patterns.
- Recommends targeted cooling measures.
- Simulates potential cooling before implementation.

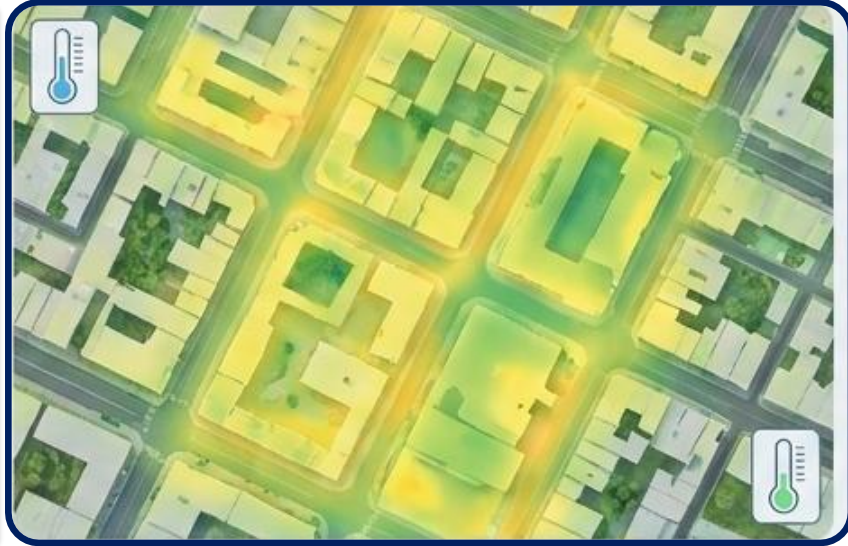
SYSTEM ARCHITECTURE

ANALYSIS PIPELINE

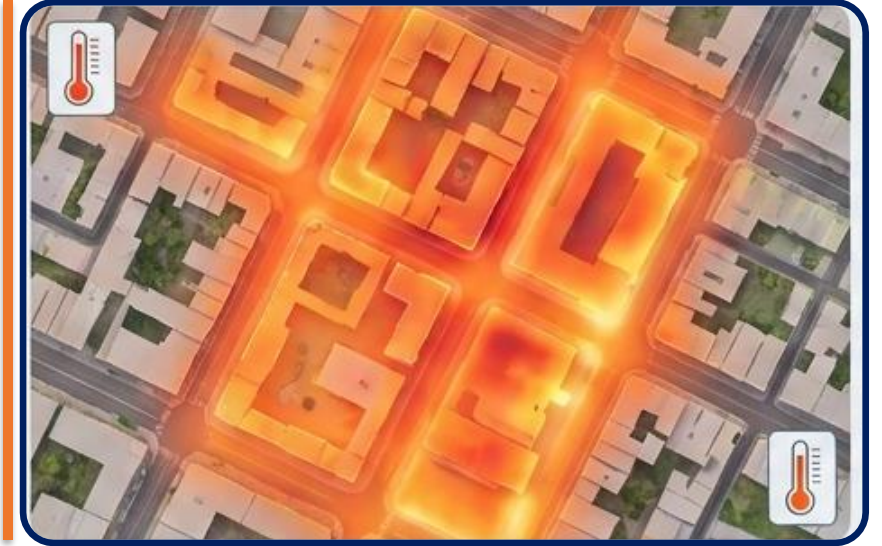
How the System Works

Upload a satellite or urban image and the system runs it through a complete AI-assisted analysis pipeline.





Mitigation applied:
+20% Canopy Cover



1. Baseline Analysis:
Select a city block to view
current heat intensity.



2. Scenario Testing: Apply
a strategy (e.g., adding
green spaces or cool roofs).



3. Actionable Results: AI
recalculates the projected
temperature drop instantly.

TECHNICAL STACK

FRONTEND

HTML • CSS •
JavaScript • Bootstrap



COMPUTER VISION

OpenCV + Segmentation Model
(candidates: SegFormer / U-Net
/ DeepLabV3)



MAPPING / GIS

Leaflet • GIS layers



BACKEND

Python • Flask



MACHINE LEARNING

Heat prediction (candidates:
Random Forest / XGBoost /
LightGBM)



VISUALIZATION

Chart.js dashboards



ANALYSIS

TECHNICAL FEASIBILITY

- Satellite & environmental data → urban analysis
- Computer vision → land-cover detection
- ML models → heat prediction
- Simulation → compare mitigation strategies
- Web dashboard → visualize results for planners

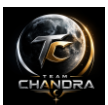
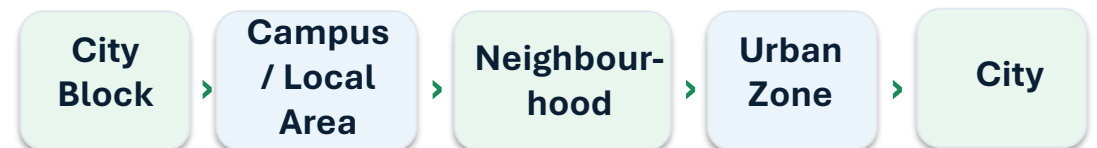
CHALLENGES

- Localized temperature data
- Image quality & resolution
- Urban variation
- Model accuracy & validation
- Computational complexity

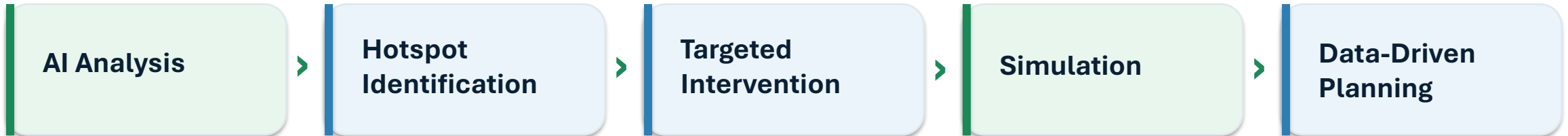
MITIGATION STRATEGIES

- Combine geospatial & environmental data
- Start with city-block analysis
- Validate using temperature/LST data
- Use modular AI/ML architecture
- Scale from blocks to cities

SCALABILITY



FROM DETECTION TO ACTION



**Targeted Green
Infrastructure
Planning**



**Optimized Water-
Based Cooling
Interventions**



**High-Albedo / Cool-
Roof Area
Identification**



**Potential Reduction
Of Heat Stress**



**Potential Reduction in
Cooling-Energy
Demand**

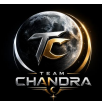
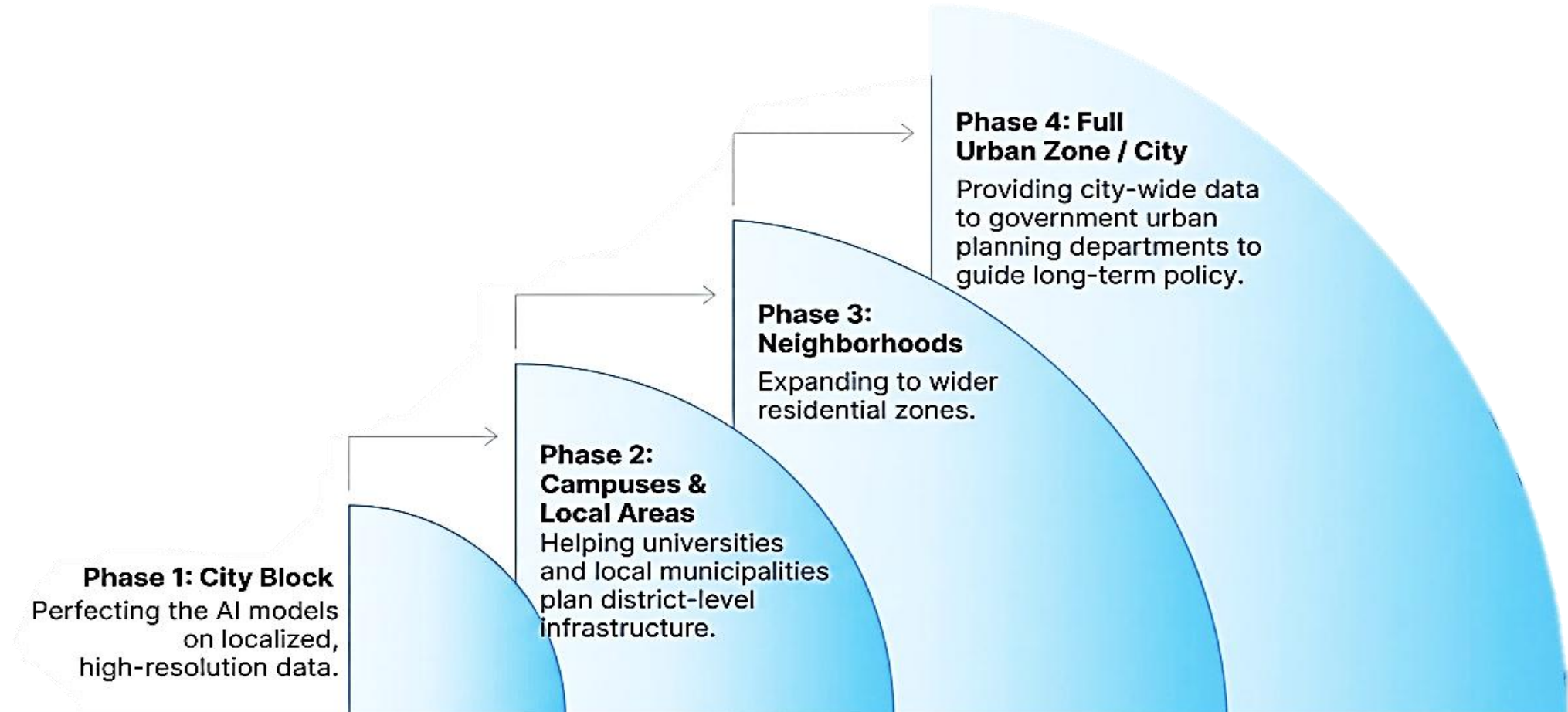


**Scalable, Data-Driven
Urban Planning**



DEPLOYMENT & SCALABILITY

Designed to scale progressively, requiring minimal structural changes as we expand the analysis area.



THANK YOU

